

Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Semester: (Spring, Year: 2025), B.Sc. in CSE (Day)

Web Engineering based Poly-network Chat System (PNCS) using Bluetooth Low Energy (BLE) for Effective Offline Communication.

Course Title: Integrated Design Project - II
Course Code: CSE-406
Section: 221-D19

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Project Report Status	
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Chapter 1

Introduction

1.1 Overview

With this project, we aim to create a messaging system using Bluetooth Low Energy (BLE) and web engineering to enable communication without the need for internet access. This system will be designed to function as a decentralized messaging network, allowing users to send and receive messages across devices directly, without relying on traditional internet-based infrastructure. The main goal is to offer an alternative means of communication, particularly during internet blackouts or government-imposed shutdowns, as recently experienced in Bangladesh. By utilizing the power of BLE and a mesh network, this system will be able to connect multiple devices over short distances, enabling people to stay connected even in challenging situations where internet or cellular services are unavailable.

1.2 Problem Domain

During recent events in Bangladesh, the government imposed widespread internet shutdowns, preventing the general population from accessing internet-based communication platforms such as Messenger [4], WhatsApp [7], and Telegram [6]. This situation highlighted several issues:

- **Disruption of Daily Communication:** People were unable to use Internet services, which affected both personal and professional communication.
- **Isolation During Crisis:** Citizens were cut off from their loved ones and communities during times of political or social uncertainty.
- **Vulnerability of Internet-based Platforms:** Internet-dependent platforms proved unreliable in emergencies where government-imposed restrictions were applied.

1.3 Motivation

To address these issues, this project proposes the use of Bluetooth Low Energy (BLE) technology to create a decentralized local communication network. Key benefits of this approach include:

- **No Reliance on the Internet:** Users can connect directly to nearby devices, bypassing internet-based platforms.
- **Continued Communication During Blackouts:** The system will allow people to stay in touch during national blackouts or emergencies
- **Decentralized Network:** The use of BLE ensures a localized network, with no need for centralized control or internet infrastructure.

This solution ensures that communication remains possible even in challenging situations where traditional networks fail.

1.4 Objectives

- **Develop a Decentralized Messaging Platform:** Use BLE and mesh networking for offline communication without centralized servers.
- **Ensure Reliable and Secure Communication:** Implement end-to-end encryption and dynamic routing to guarantee secure, reliable message delivery.
- **Enhance Accessibility and User Experience:** Create a modern web-based interface that is intuitive, user-friendly, and compatible with various devices.
- **Create a Cost-Effective Solution:** Utilize open-source tools and ensure compatibility with existing devices to reduce costs and hardware dependencies.
- **Promote Scalability:** Design a modular architecture that supports growth with user demand, enabling dynamic node discovery and self-healing networks.
- **Foster Decentralization:** Eliminate centralized control to provide autonomy, privacy, and resistance to censorship or failures.

Chapter 2

Literature Review

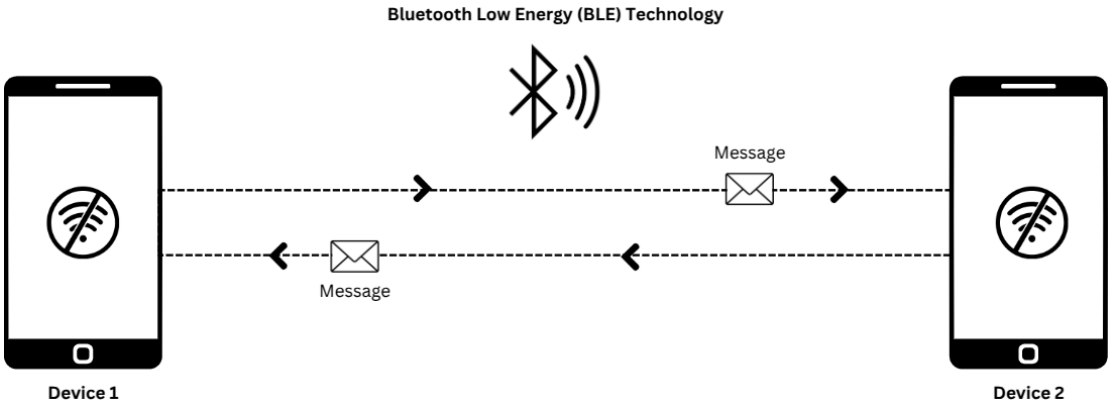
In recent years, device-to-device (D2D) communication has gained significant attention as a means to improve network efficiency and enable seamless connectivity in environments where traditional internet infrastructure is unavailable. The advent of the Internet of Things (IoT) has further fueled this development by providing smart, interconnected devices that rely on low-latency communication for efficient operation. This literature review focuses on the core concepts of intelligent D2D communication, including technologies such as Wi-Fi Direct and Bluetooth, as well as the role of such technologies in building an offline messaging system, as proposed in our project.

Key Research and Technologies

Bello and et a [1] explored intelligent D2D communication in IoT, emphasizing how devices can form direct communication links without centralized control or internet access. Their study highlights the potential for D2D communication to reduce latency and power consumption while improving data transfer efficiency in dense IoT ecosystems. This research aligns with our project's goal of leveraging local device networks for seamless messaging without internet connectivity. In addition, Camps-Mur et al. [3] provided an overview of Wi-Fi Direct as a promising D2D communication protocol. Their work demonstrated the efficiency of Wi-Fi Direct in enabling high-speed data transfers between devices without the need for a WiFi hotspot or infrastructure, making it an ideal candidate for local communication in offline environments. This technology serves as a foundation for our project's communication protocol. Further research by Bello et a [2] examined the interoperation of D2D communication technologies in the IoT, particularly focusing on Bluetooth, Zigbee, and IPv6. Their findings underscore the challenges in achieving interoperability between different communication standards and suggest potential solutions for enhancing cross-device communication, which is a critical aspect of our project. Finally, Gismalla et a [5] provided an overview of D2D communication within IoT ecosystems, emphasizing the importance of developing secure and efficient communication protocols to manage the increased data traffic from billions of interconnected devices.

Chapter 3

Methodology



3.1 SRS (Software Requirement Specification)

The Software Requirement Specification (SRS) document defines the functional and non-functional requirements of the system. It provides a clear understanding of the project goals, system behavior, and constraints.

The SRS for our BLE Messaging System includes:

- **Functional Requirements:**
  - User Authentication and Authorization.
  - Message Sending, Receiving, and Storing.
  - Encryption and Secure Communication.
  - Data Encryption
  - Cross-platform Compatibility
- **Non-Functional Requirements:**

- Performance Efficiency.
- Data Confidentiality.
- System Availability and Scalability.
- Maintainability
- Usability

3.2 SDLC (Software Development Life Cycle)

The system follows the Agile Model of the Software Development Life Cycle (SDLC). The phases include:

Table 1: Selecting SDLC models for PNCS

Priority	Criteria	Waterfall	V-shape	Iterative	Spiral	Agile	Prototype
3	Well-known requirements	Yes	Yes	No	No	No	No
3	Technological knowledge	Yes	Yes	Yes	No	No	No
6	Efficiency with BLE technology	No	No	No	Yes	Yes	No
4	Risk analysis for decentralized systems	No	No	Yes	Yes	Yes	No
5	User testing for BLE applications	No	No	Yes	No	Yes	Yes
5	Security in communication	No	No	Yes	Yes	Yes	Yes
6	Dependability	No	Yes	Yes	Yes	Yes	Yes
4	Time-consuming development	Yes	Yes	No	No	No	Yes
5	Authorization	No	No	Yes	Yes	Yes	No
Total: 39	Overall score	10	16	28	26	31	25

3.3 DFD (Data Flow Diagram)

A Data Flow Diagram (DFD) represents the flow of data within the system. The DFD explains how data moves between users, processes, and storage.

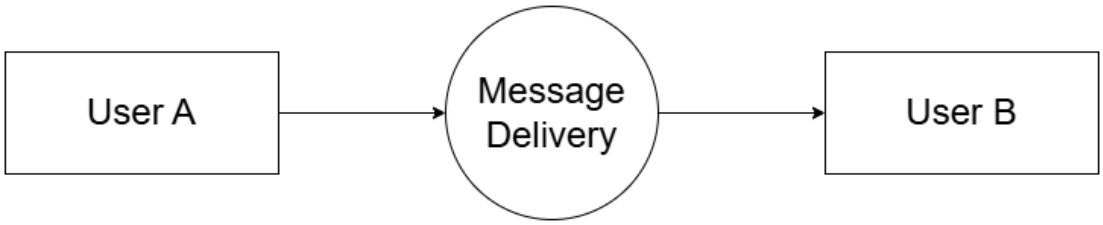


Figure 3.1: Data Flow Diagram Level:0 of PNCS

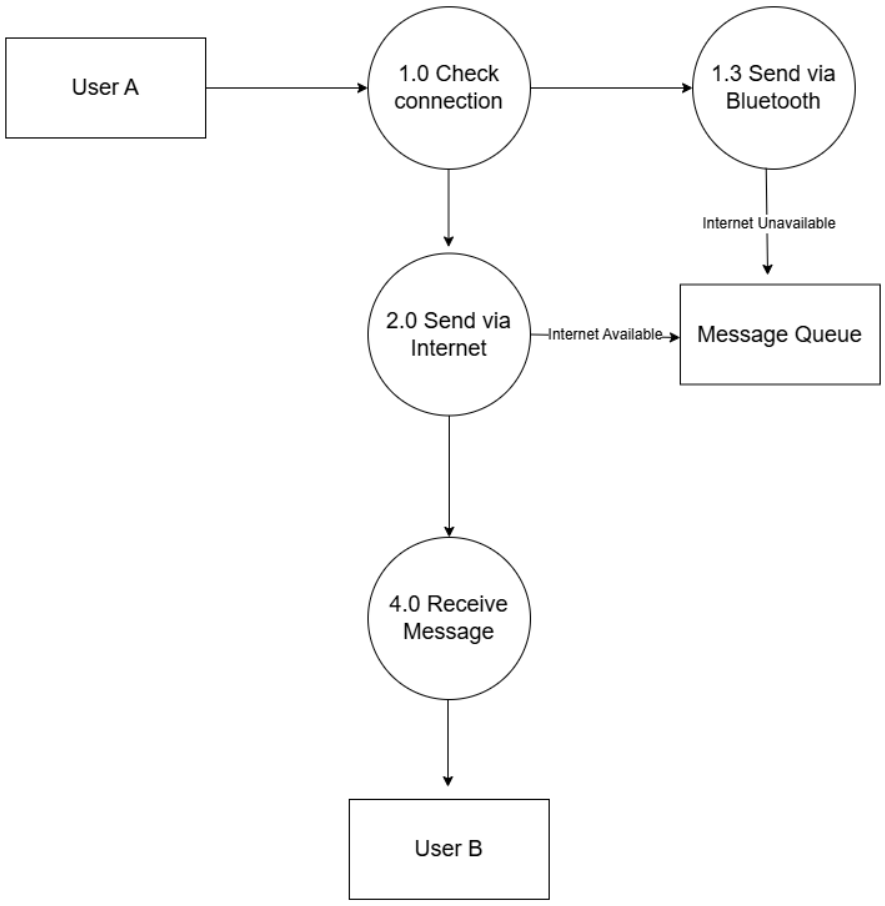


Figure 2.1 : Level 1 DFD for PNCS

Figure 3.2: Data Flow Diagram Level:1 of PNCS

3.4 UML Use Case Diagram

The Use Case Diagram provides an overview of the interactions between the system and its users. It highlights the various functionalities accessible to the actors.

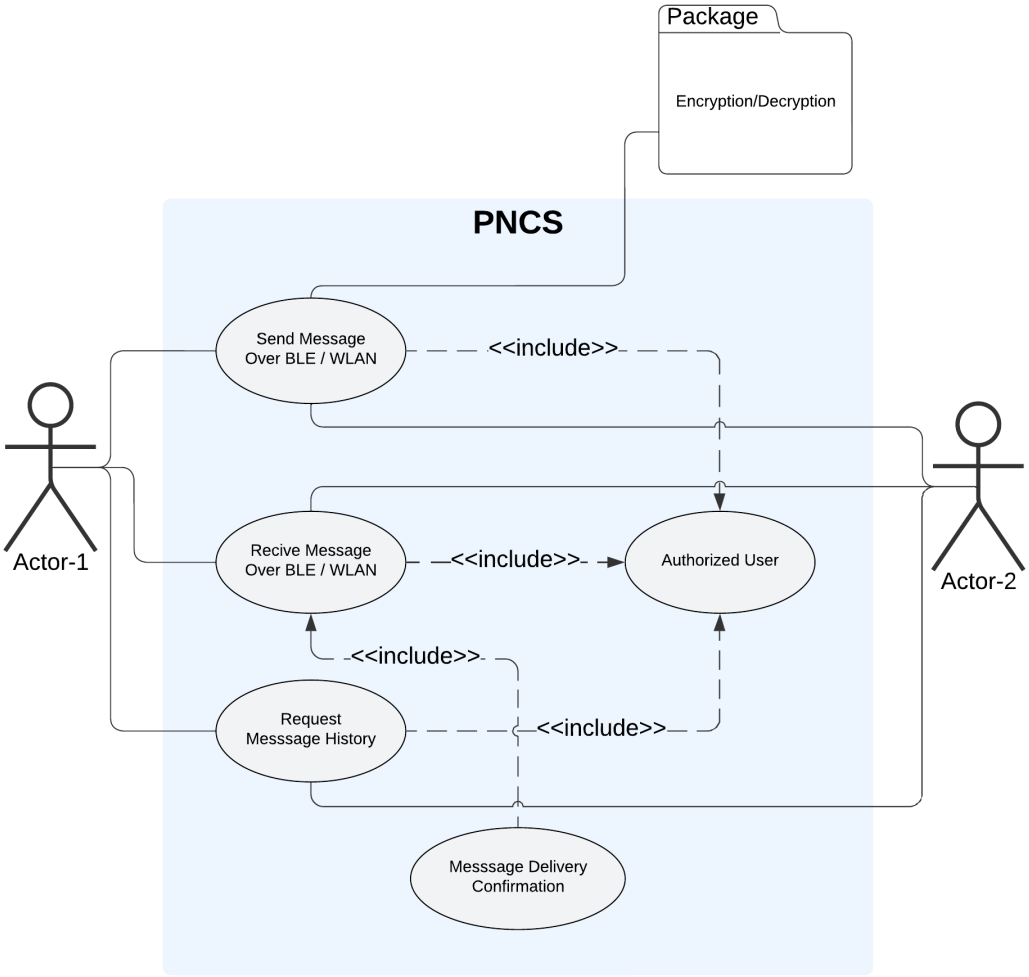


Figure 3.3: UML Use Case Diagram of PNCS

3.5 UML Sequence and Communication Diagram

The Sequence Diagram depicts the flow of messages and interactions between objects in a specific sequence over time. The Communication Diagram focuses on the relationship between objects, showing how messages are exchanged to achieve functionality.

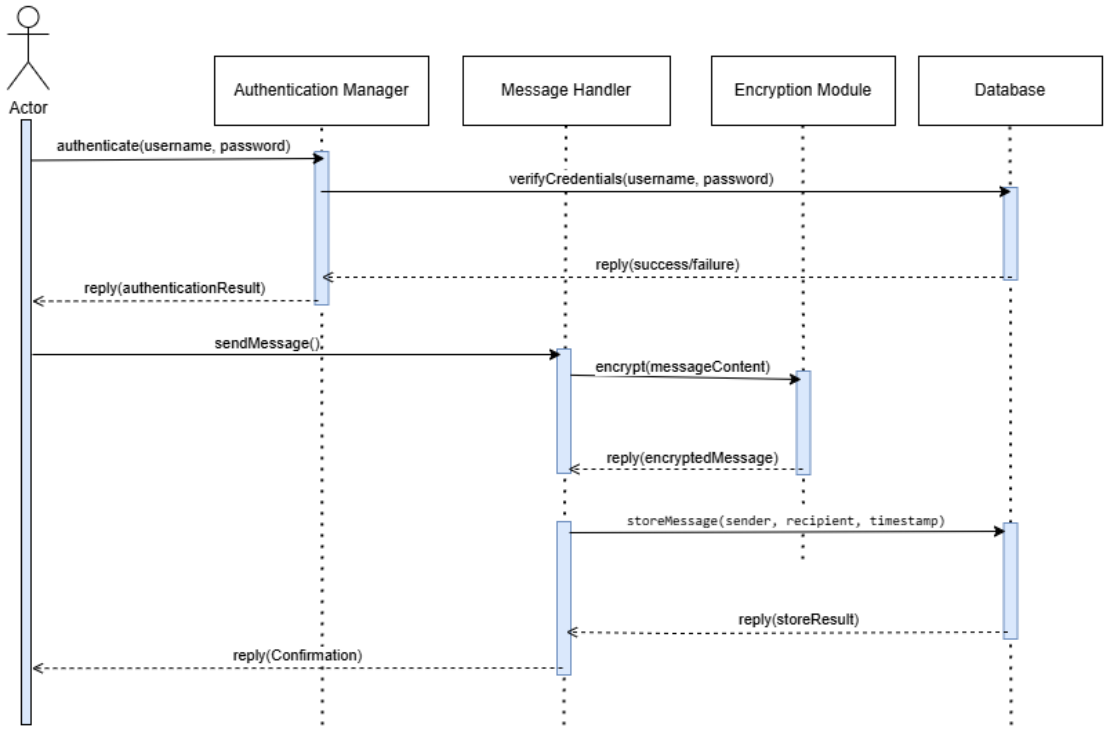


Figure 3.4: UML Sequence and Communication Diagram for PNCS

3.6 UML Class Diagram

The Class Diagram illustrates the static structure of PNCS Messaging. It defines the system's classes, attributes, methods, and relationships.

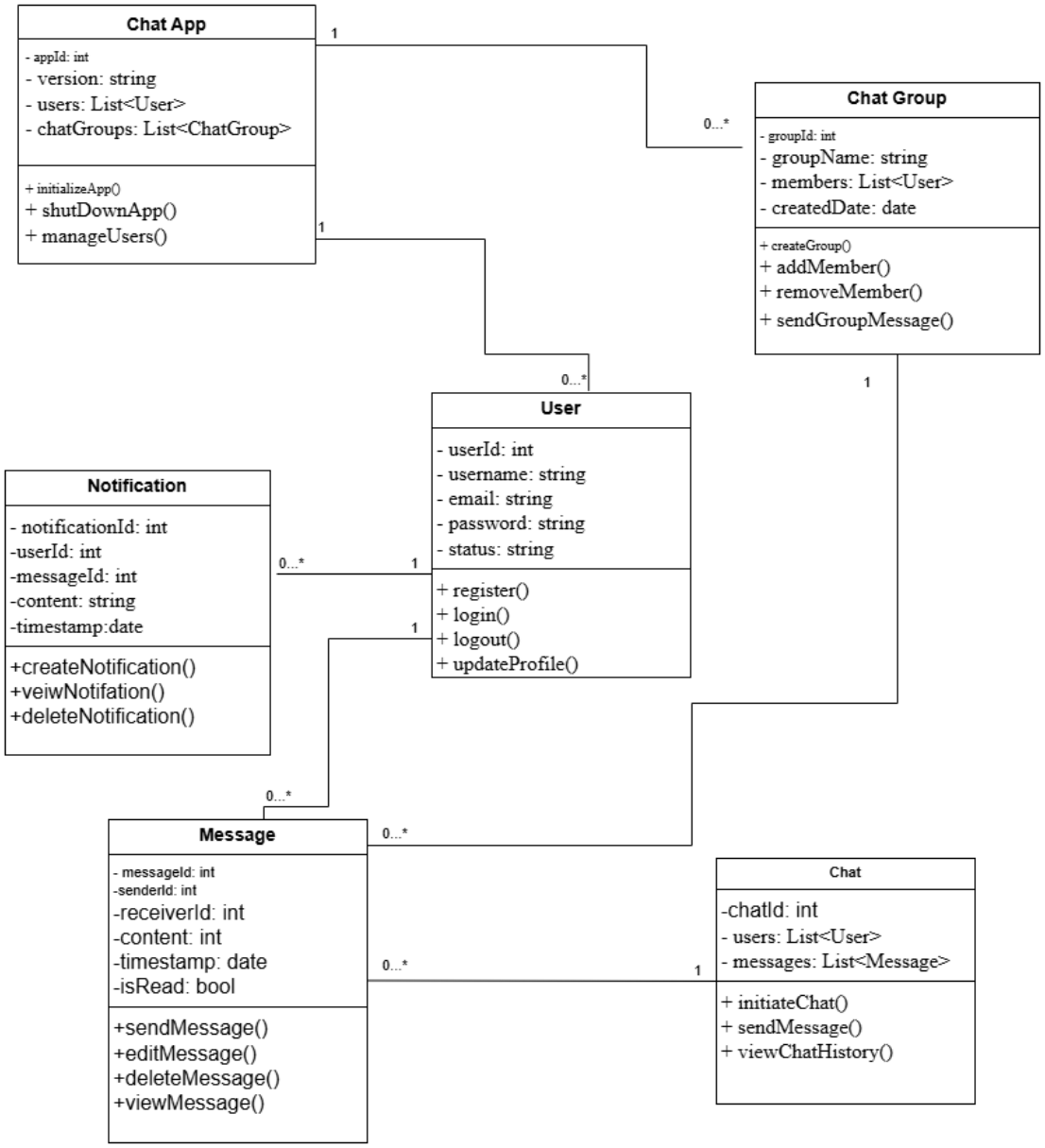


Figure 3.5: UML Class Diagram of PNCS

3.7 Gantt chart

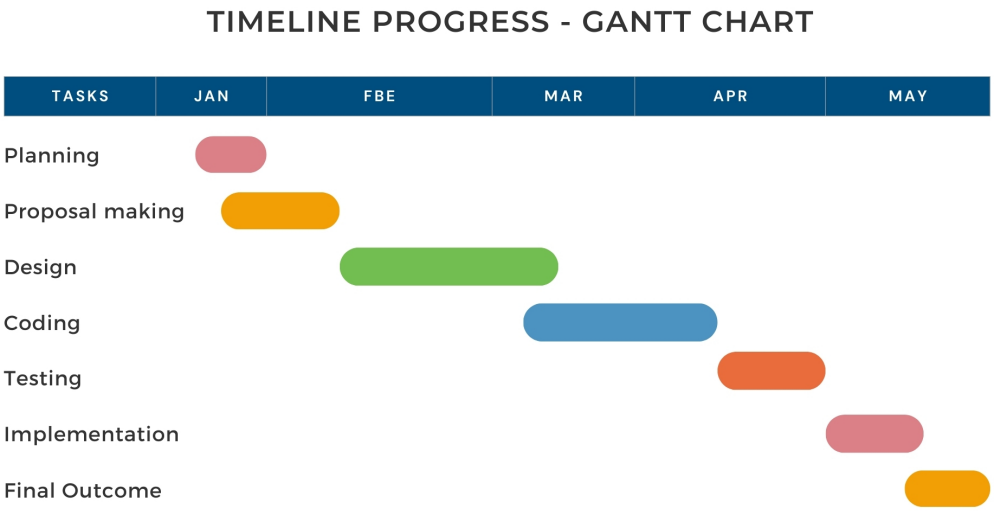


Figure 3.6: Gantt chart of PNCS

Chapter 4

System Architecture

4.1 Introduction

The Poly-network Chat System (PNCS) is a web-engineering-based communication platform designed for effective offline messaging using Bluetooth Low Energy (BLE) and Wi-Fi Direct. The system enables seamless communication between users without requiring an internet connection. The Admin Panel is a critical component of PNCS, allowing administrators to manage users, messages, security, and system logs. This document focuses on both the High-Level Design (HLD) and Low-Level Design (LLD) of PNCS, detailing its internal components and functionalities. The HLD provides an overview of the system architecture, key modules, and interactions between components

to ensure scalability, security, and efficiency. It defines the overall system structure, in- cluding major components such as the communication framework, user management,

and data flow. The LLD delves into the finer details of the system, covering Message Processing, Communication Modules (BLE and Wi-Fi), Security, Notification System,

and Database Management. The paper explores the design, methodology, and implementation to ensure that the system is robust and optimized for offline communication.

4.2 High Level Design (HLD)

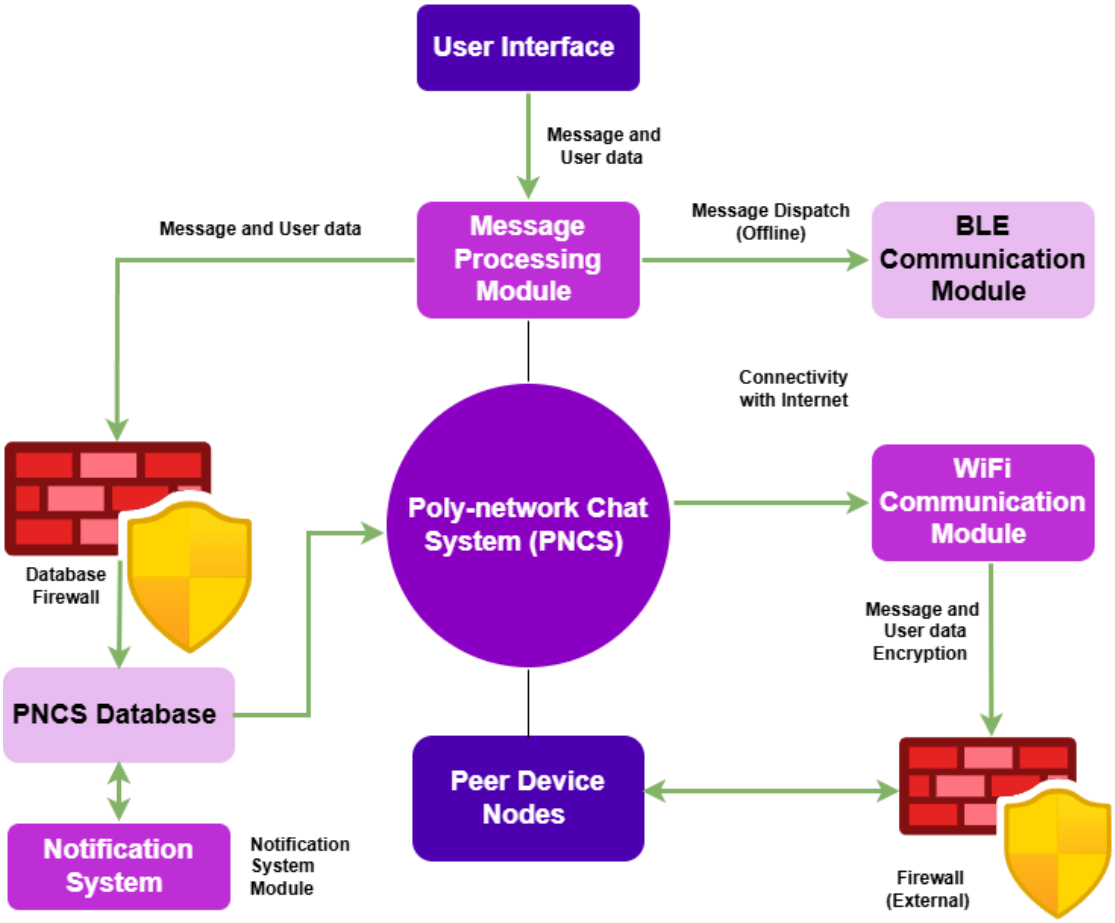


Fig:High-Level System Architecture

Figure 4.1: High level (HLD) architecture (abstract view) of the system of PNCS

4.3 Low Level Design (LLD)

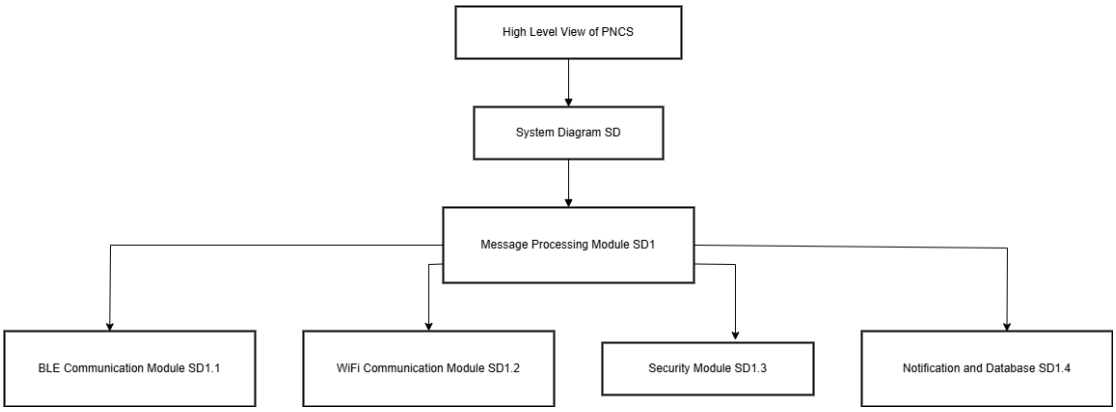


Figure 4.2: Trends from High-Level View to Low-Level View

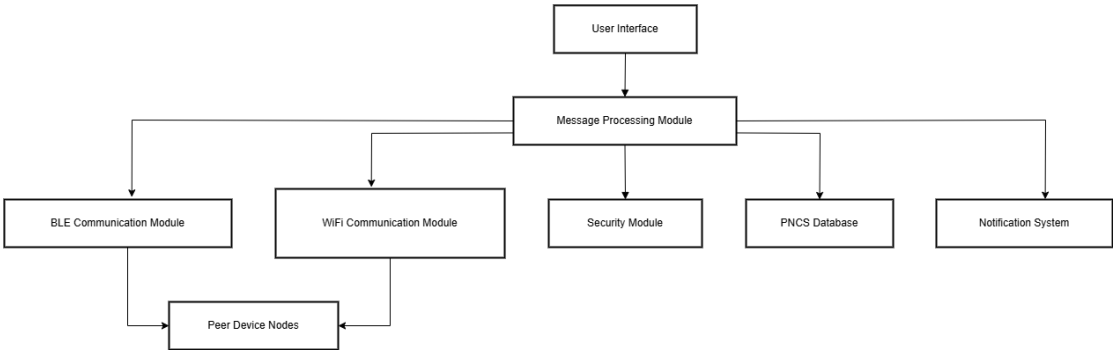


Figure 4.3: SD - Low-Level View of the System Architecture

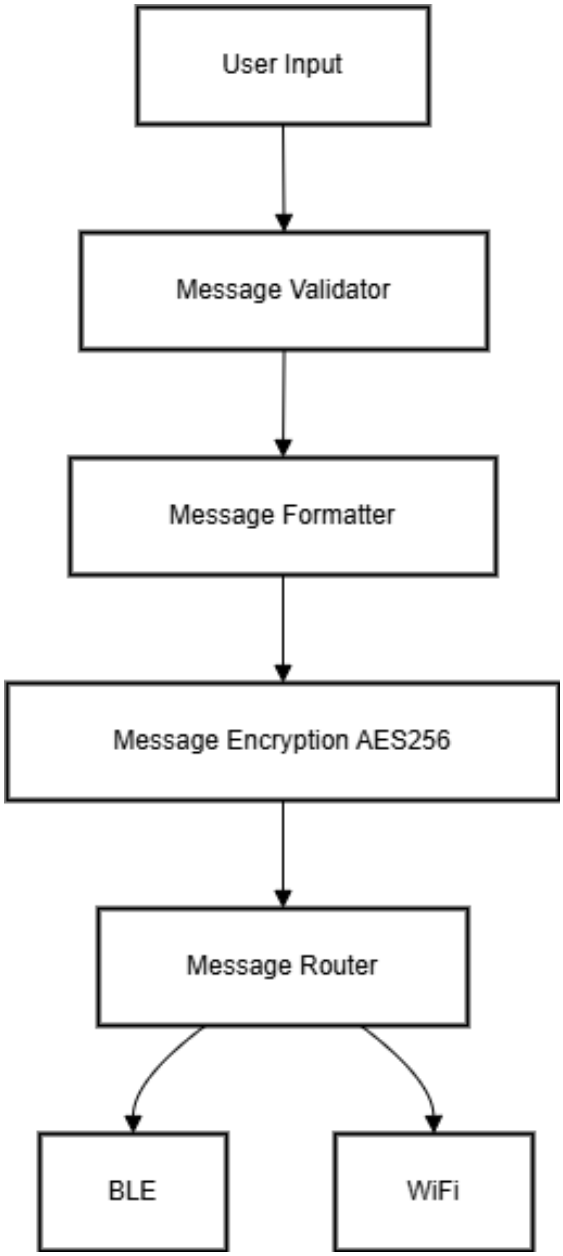


Figure 4.4: SD1 - Low-Level View of Message Processing Module

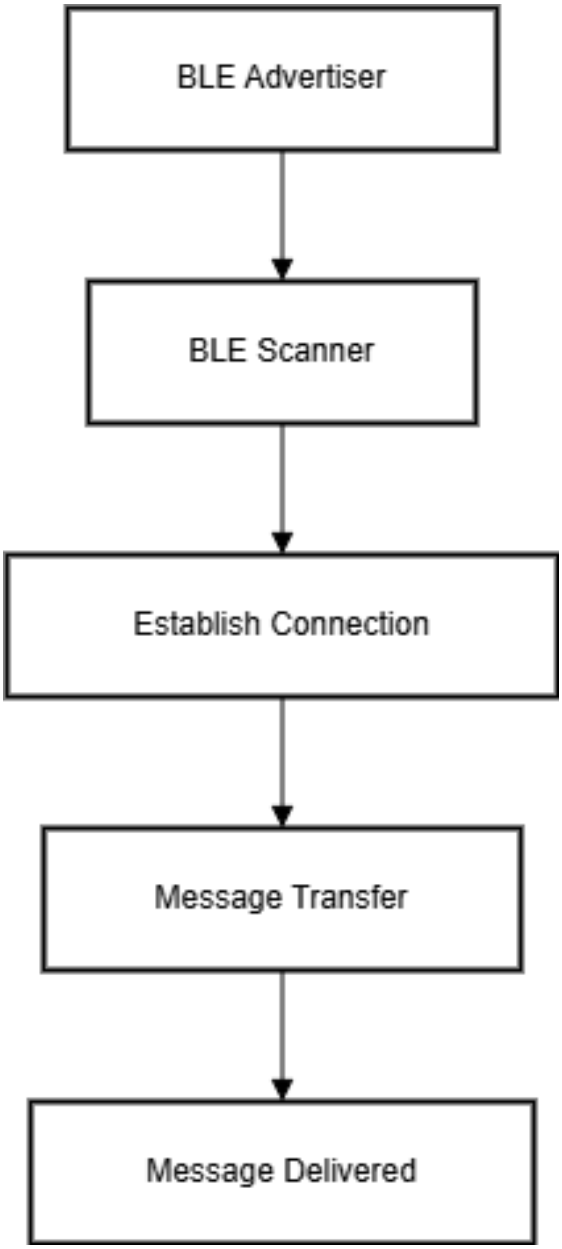


Figure 4.5: SD1.1 - Low-Level View of BLE Communication Module

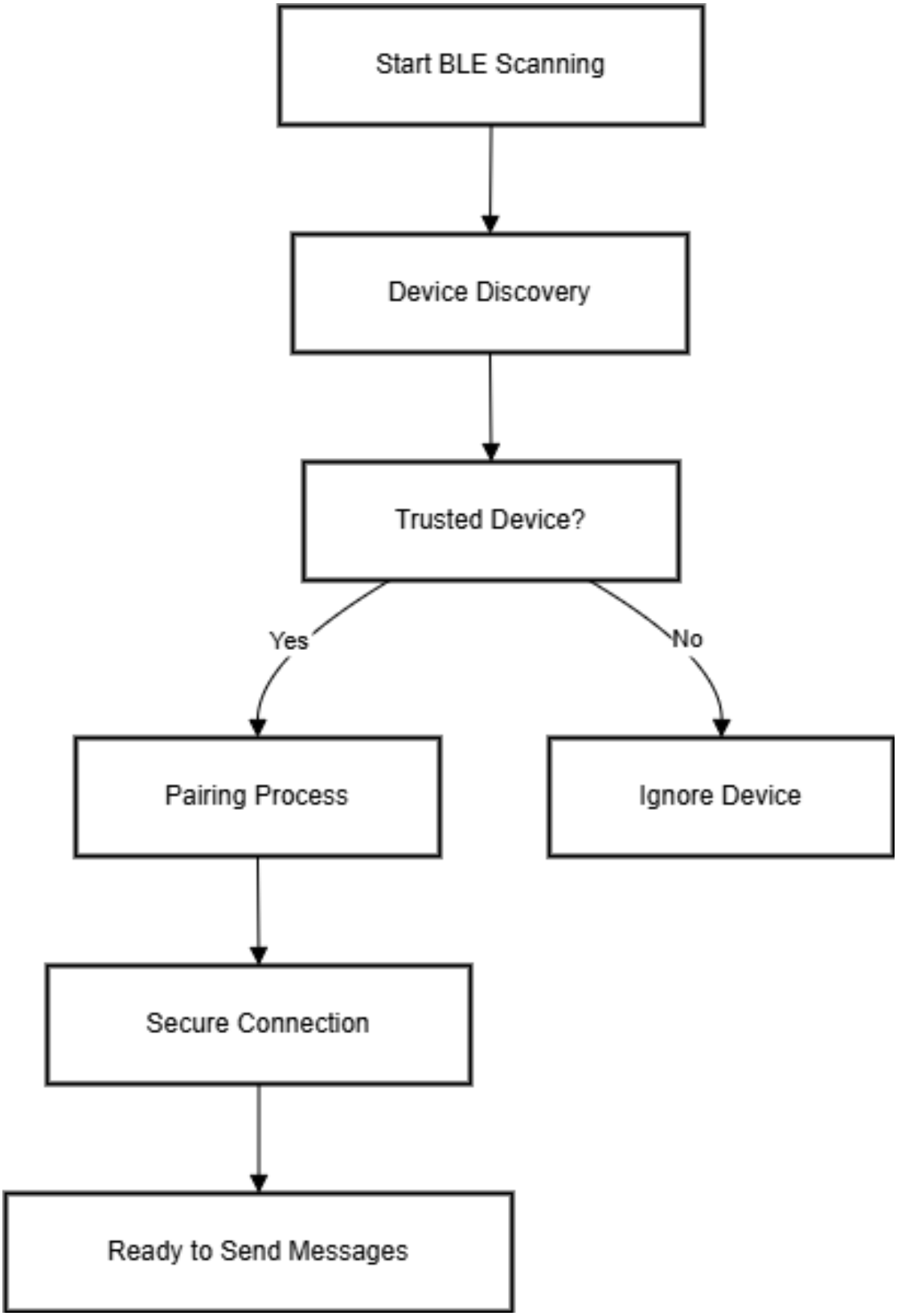


Figure 4.6: SD1.1.1 - Low-Level View of BLE Device Discovery and Pairing

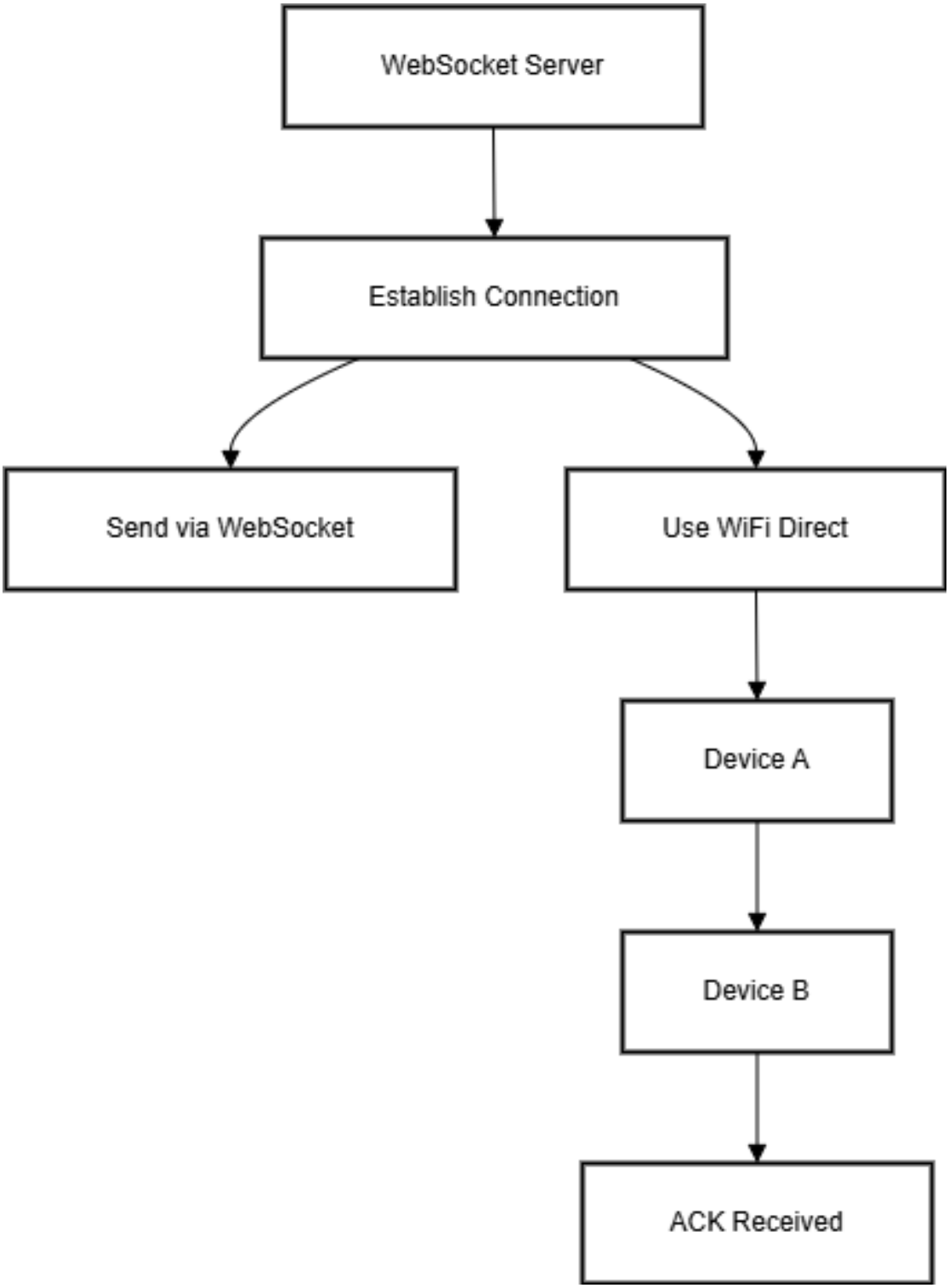


Figure 4.7: SD1.2 - Low-Level View of WiFi Communication Module

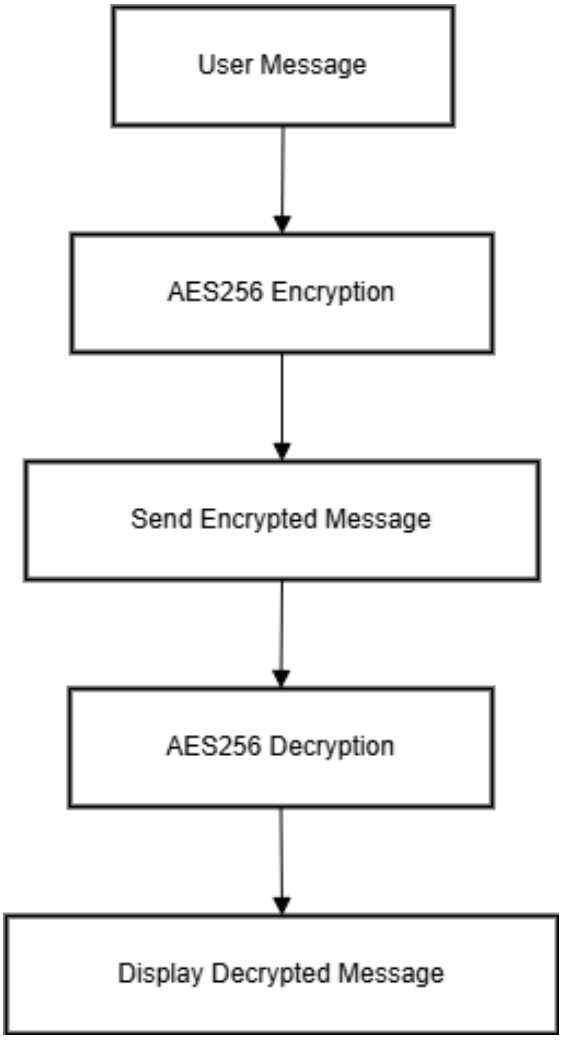


Figure 4.8: SD1.3 - Low-Level View of Security Module

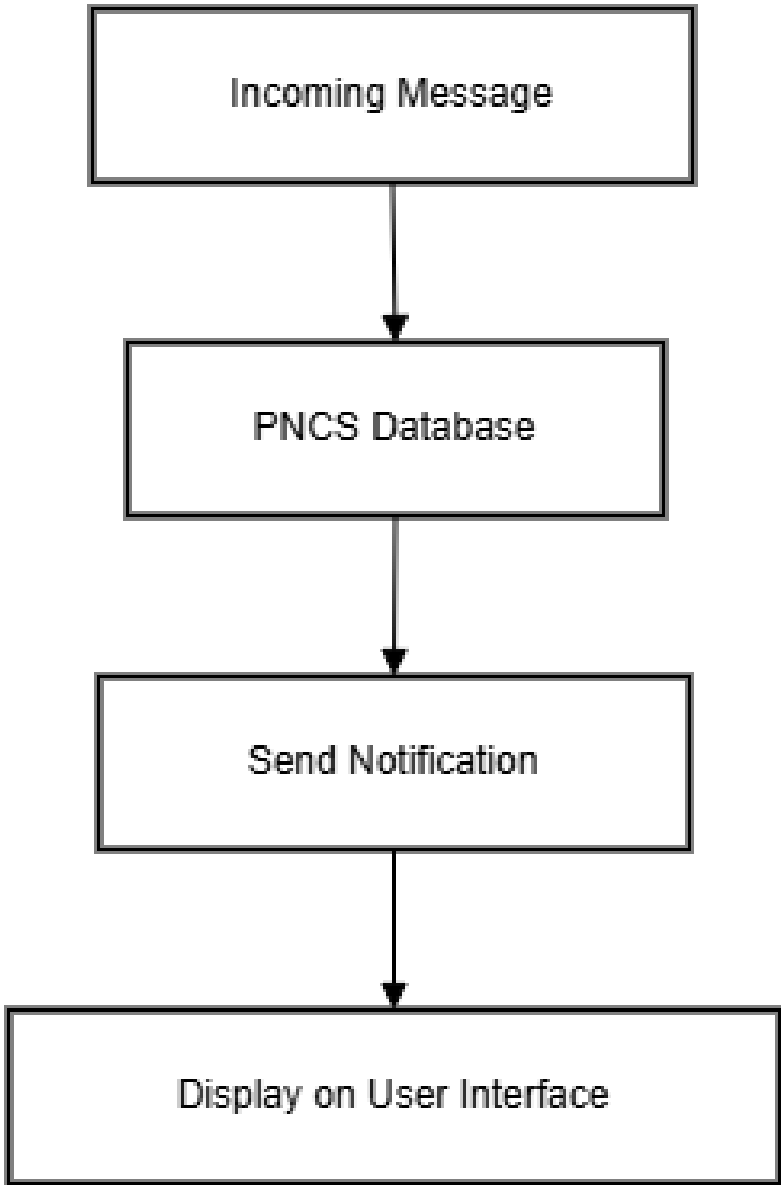


Figure 4.9: SD1.4 - Low-Level View of Notification System and Database Integration

4.4 ER Diagram

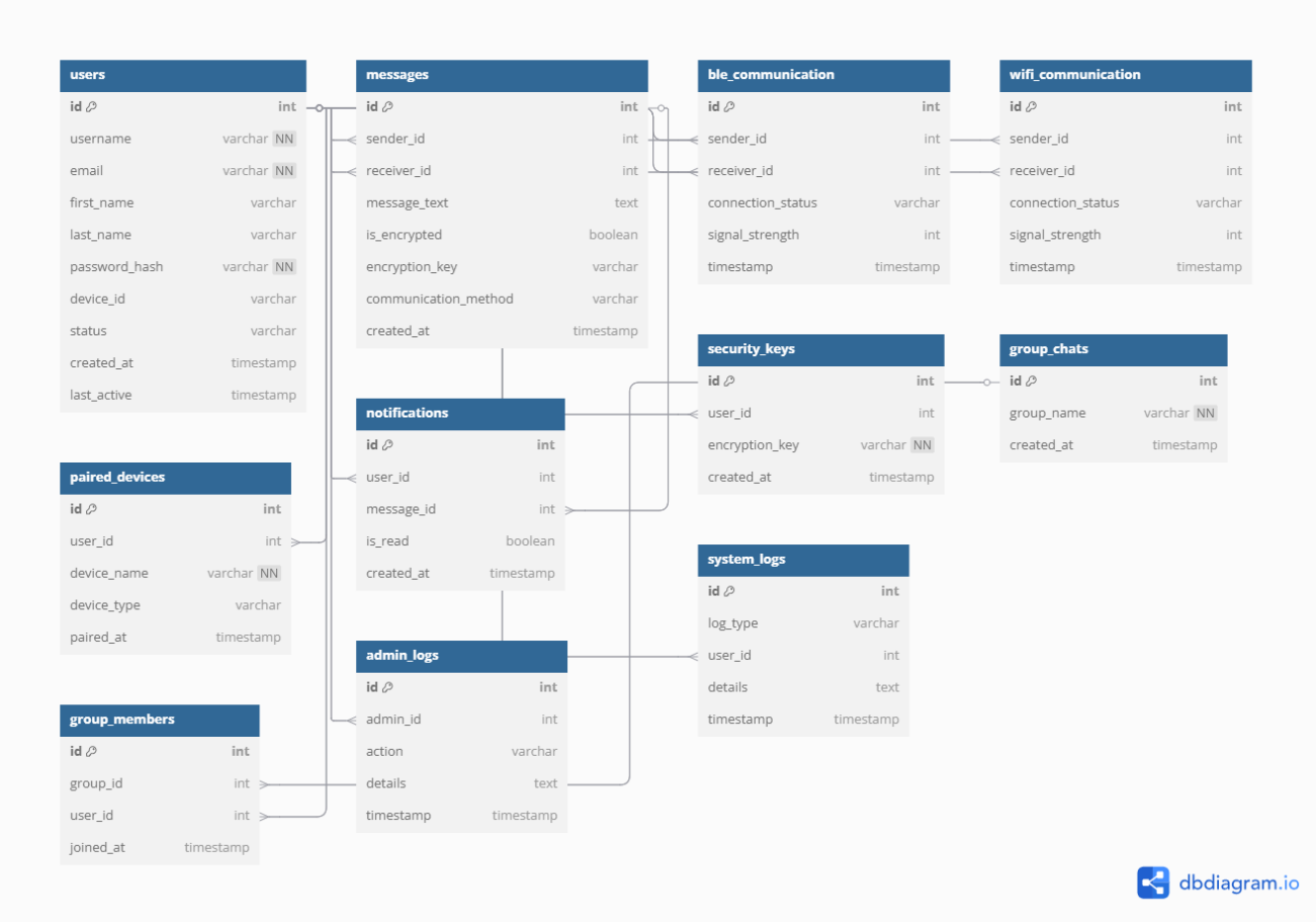


Figure 4.10: ER Diagram of PNCS

Chapter 5

UX-UI Design

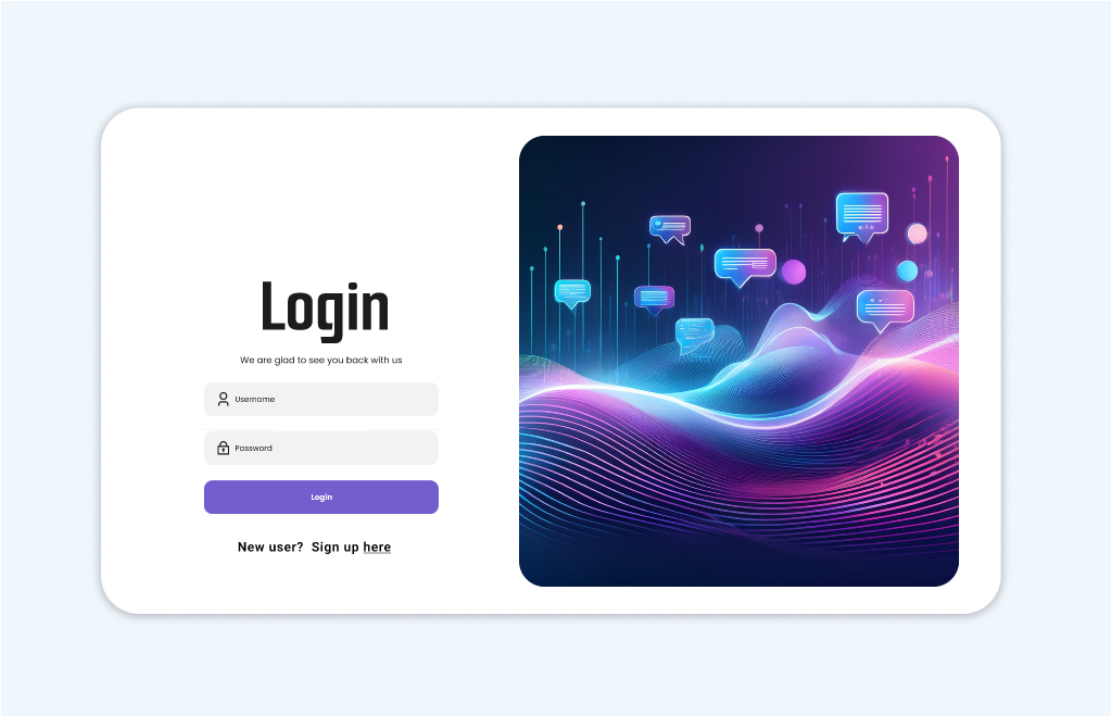


Figure 5.1: Login page of PNCS

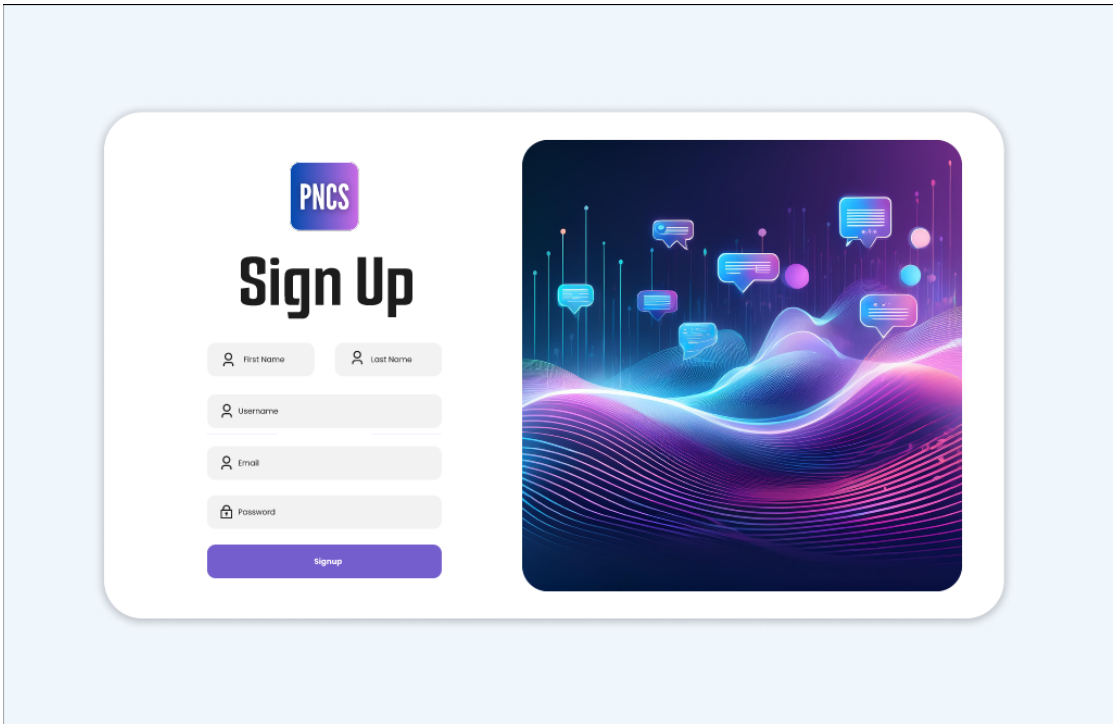


Figure 5.2: Signup page of PNCS

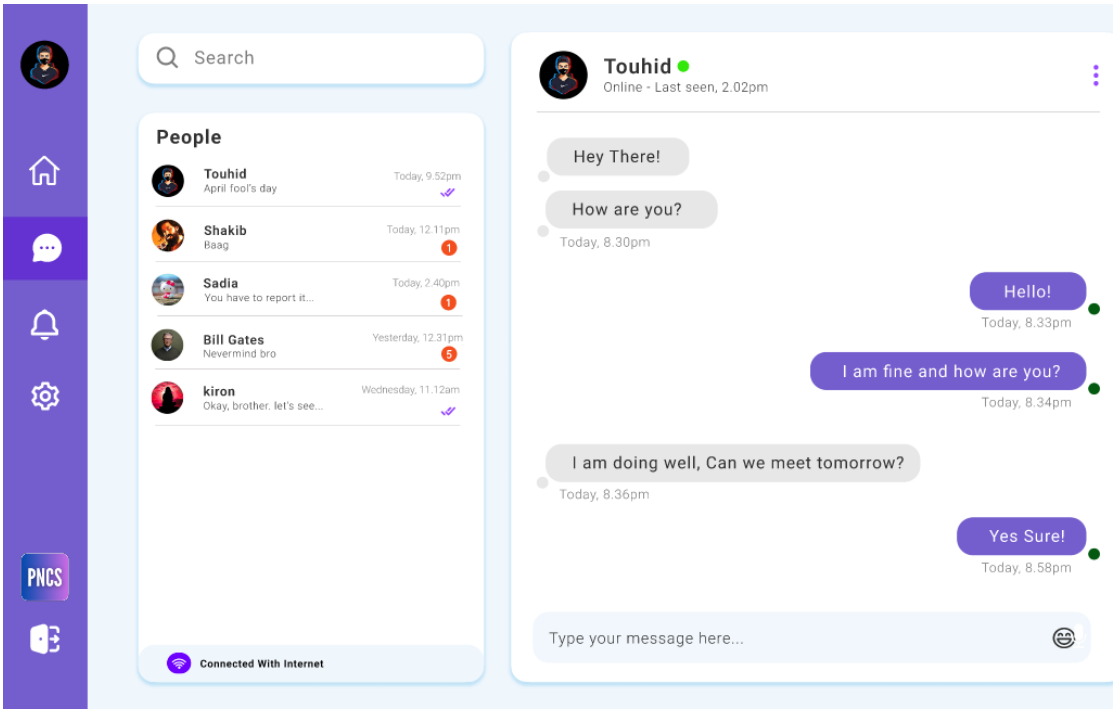


Figure 5.3: Home page of PNCS

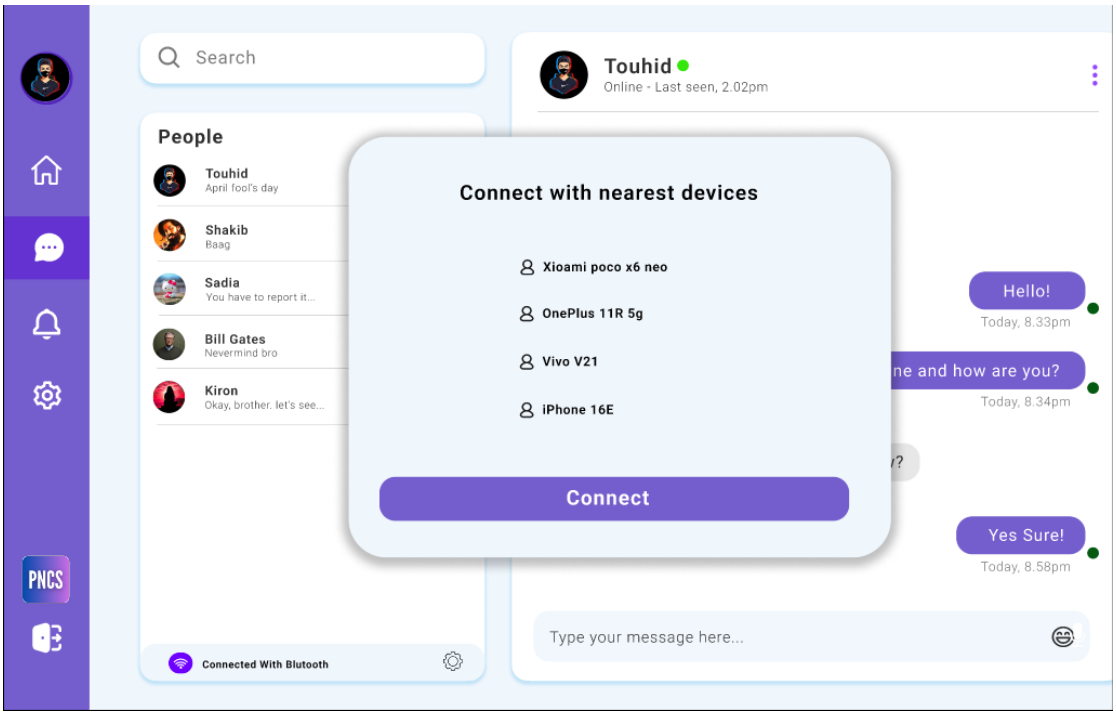


Figure 5.4: Bluetooth settings of PNCS

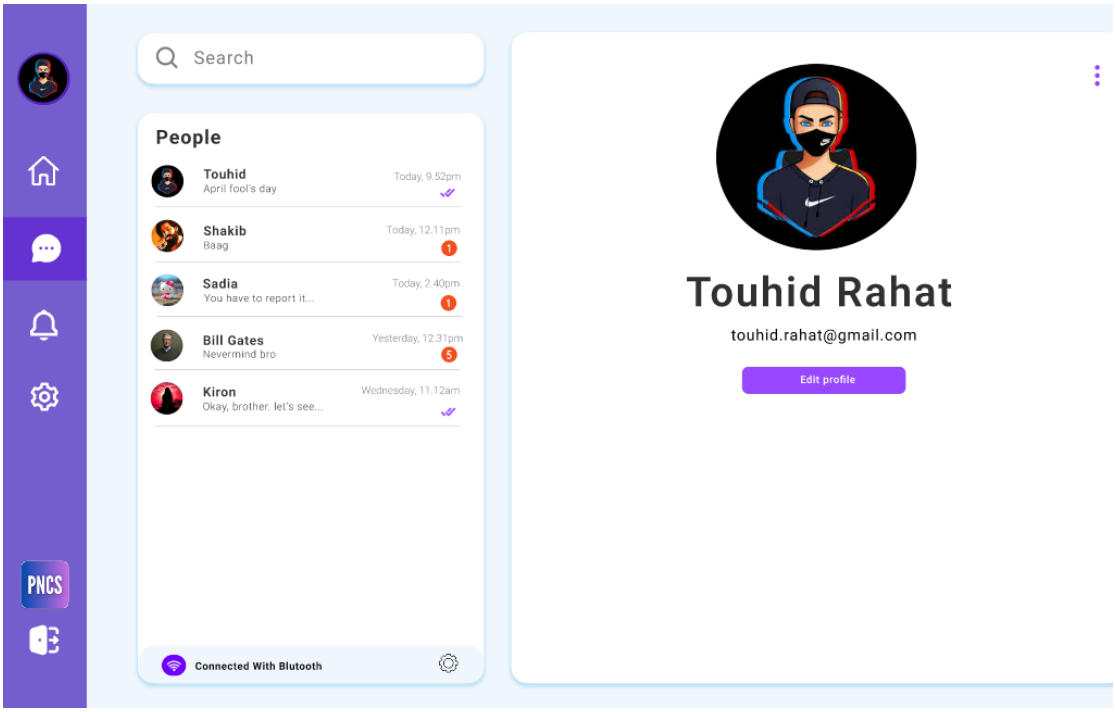


Figure 5.5: Profile page of PNCS

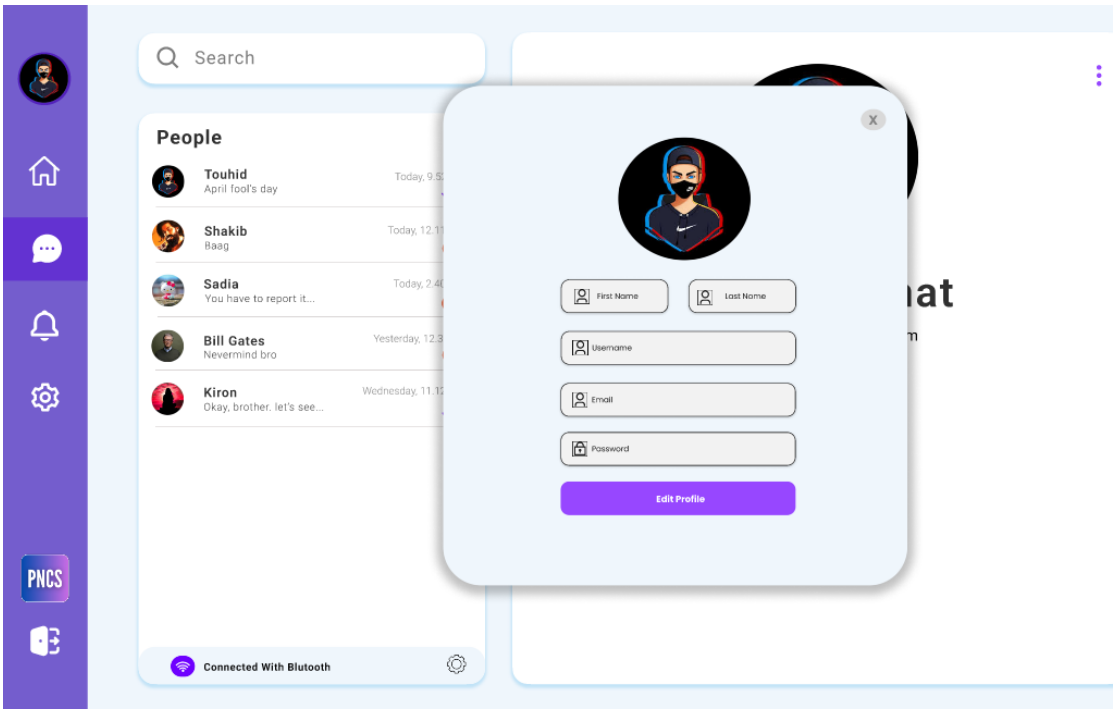


Figure 5.6: Edit profile page of PNCS

Chapter 6

Test Cases

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Valid sign up details	Enter valid details and submit	Valid username, email, password, user image upload	Account created successfully	Account created successfully, navigate to home page	Passed	4	4
Missing required fields	Submit with empty fields	Empty username, email, password	Error message for required fields	Password is required or Email is required	Passed	3	3
Invalid email format	Enter invalid email	Invalid email address	Error message for invalid email	Firebase: Error (auth/invalid-credential)	Passed	2	2
Weak password	Enter weak password	Simple password (e.g., 12345)	Error message for weak password	Password must contain at least one uppercase letter, one lowercase letter, one digit, and one special character.	Passed	2	2

Figure 6.1: Signup Page

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Valid login	Enter valid credentials	Email: valid, Pass: valid	Login successful, redirect to Home	Login successful, redirect to Home	Passed	5	5
Invalid login	Enter wrong password	Email: valid, Pass: wrong	Show 'Incorrect password' error	Show 'Incorrect password' error	Passed	4	4
Empty fields	Submit without email/password	Empty email and password	Show 'Fields cannot be empty' error	Show 'Fields cannot be empty' error	Passed	3	3
Invalid email format	Enter wrong email format	Email: abc123	Show 'Invalid email format'	Show 'Invalid email format'	Passed	2	2

Figure 6.2: Login Page

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Send message	Type and send message	Message: 'Hello'	Message appears in chat window	Message appears in chat window	Passed	6	6
Empty message	Click send with no text	Empty input	Do not send; show warning	Do not send; show warning	Passed	3	3
Logout button	Click logout	Click 'Logout'	Redirect to login page	Redirect to login page	Passed	4	4

Figure 6.3: Home Chat

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Pairing available device	Click available device	Device found	Device paired successfully	Device paired successfully	Passed	3	3
No device found	Open Bluetooth settings	No Bluetooth devices available	Show message: 'No devices found'	Show message: 'No devices found'	Passed	2	2

Figure 6.4: Bluetooth Settings Page

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
View profile	Login and open profile	User is logged in	Display profile details	Display profile details	Passed	2	2
Click Edit	Click on Edit Profile	Click 'Edit'	Redirect to Edit Profile page	Redirect to Edit Profile page	Passed	3	3

Figure 6.5: Profile Page

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Update profile	Submit valid fields	New username and photo	Profile updated successfully	Profile updated successfully	Passed	2	2
Submit empty fields	Click submit with empty fields	Empty fields	Show validation error	Show validation error	Passed	2	2

Figure 6.6: Edit Profile Page

Criteria	Action	Input (Test Case)	Expected Output	Actual Output	Test Result	Times Tested	Times Passed
Responsive design	Open on mobile/tablet	Small screen	Layout adapts responsively	Layout adapts responsively	Passed	2	2
Session timeout	Stay idle 15 mins	Idle time: 15 mins	Auto logout and redirect to login	Auto logout and redirect to login	Passed	1	1

Figure 6.7: Additional Test Cases

Chapter 7

Conclusion

The proposed platform offers an innovative solution for connecting volunteers with local organizations through geolocation technology, ensuring faster and more effective collaboration. By leveraging modern web engineering, the system is designed to be scalable, user-friendly, and robust, fostering community engagement and enhancing resilience during emergencies. Additionally, it supports sustainable local development by empowering non-profits and encouraging greater participation in social initiatives. Overall, the platform bridges the gap between volunteers and opportunities, promoting a culture of active involvement and contributing to stronger, more resilient communities.

7.1 Discussion

- **Enhancing Connectivity through Geolocation Technology:**The platform effectively utilizes geolocation services to bridge the gap between volunteers and organizations, ensuring that opportunities are matched based on proximity. This feature enables faster responses during emergencies and ensures that resources are allocated where they are needed most.
- **Scalability and Usability through Web Engineering:** By incorporating modern web technologies, such as scalable frameworks and cloud-based infrastructure, the platform offers a responsive and user-friendly experience. These design choices ensure that the system can handle high user traffic and remain accessible to a wide audience, including diverse organizations and volunteers.
- **Promoting Community Engagement and Resilience:** The platform fosters a culture of social responsibility by simplifying the process of finding and engaging in volunteer opportunities. It empowers non-profits and local initiatives, encourages sustainable development, and strengthens community resilience by supporting collaborative efforts during critical situations.

7.2 Limitation

Here are some limitations for this project.

- **Dependence on Internet Connectivity:**The platform requires reliable internet access, which may exclude users in remote or underprivileged areas with limited connectivity.
- **Accuracy of Geolocation Data:** The effectiveness of the system relies heavily on the precision of geolocation data, which can be affected by device limitations or environmental factors.
- **Data Privacy and Security Concerns:** Handling sensitive user information, such as location and personal details, poses risks if proper security measures are not implemented.
- **Limited Reach in Non-Digital Communities:** Communities with low digital literacy or access to technology may face challenges in utilizing the platform effectively.
- **Initial Implementation and Adoption Challenges:** Gaining trust and adoption from volunteers and organizations, especially in regions unfamiliar with similar platforms, may require significant outreach and education efforts.

7.3 Future Work

Future development of the Geolocation-Driven Volunteer Matching Platform . we will focus on overcoming its current limitations and enhancing its applicability:

- Integrating AI-based algorithms for more precise volunteer matching.
- Expanding multilingual support to cater to diverse user bases. Enhancing mobile application features for better accessibility.
- Implementing offline functionality for limited connectivity areas.
- Incorporating advanced data analytics for insights into volunteering trends.
- Strengthening data privacy measures with end-to-end encryption.
- Collaborating with more non-profits and local organizations globally.
- Adding gamification features to incentivize volunteer participation.

The proposed platform has significant potential for growth and innovation. Future enhancements will focus on leveraging advanced technologies such as AI for improved matching, expanding accessibility through multilingual support and offline functionality, and strengthening data privacy and security. Additionally, incorporating data analytics, fostering global partnerships, and introducing gamification features will further enhance user engagement and the platform's impact. These advancements will ensure the system continues to evolve, effectively supporting communities and promoting sustainable development worldwide.

References

[1] Oladayo Bello and Sherali Zeadally. Intelligent device-to-device communication in the internet of things. *IEEE Systems Journal*, 10(3):1172–1182, 2023.

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